

Gil Jorge

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Portugal
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EXPERIENCE AND EDUCATION

Aqua Hotels, Sales/Marketing, Lagoa— Paid *Internship*

July 2022

- Developed automated accounting data analysis workflow producing daily dashboards. Managed website's chatbot (client service) and reservations.

Champalimaud Center for the Unknown, Lisbon— *Internship*

Jan 2026-present

- Designing algorithms to temporally align independently-clocked behavioral and neural recordings when no working hardware reference signal exists — engineering and statistically evaluating signal-processing features, then building a spike-timing inference method with iterative outlier rejection and multi-unit fusion (sub-20ms precision).
- Built a batched, single-pass processing pipeline for ~80GB-per-file electrophysiology recordings across 8 probe sessions.

Instituto Superior Técnico, Lisbon— *BSc Physics Engineering*

Sep 2023 - July 2026 GPA out of 20: Year 1- 16; Year 2- 15; Year 3- 17

- Did 6 extra courses, 3 at MSc level (Abstract Algebra, Statistical Physics, Knowledge Representation and Reasoning, Cryptography and Security Protocols and Mathematics for Machine Learning). Other courses: Deep learning of MSc CS 17/20, 19/20 for presentation on Quantum Information and Computation, best grade in the cohort for project in the Quantum Photonics Lab (19/20).
- **One of 7 students** of my cohort to earn a spot at oral exam in calculus (went in multivariable calculus, calculus 2).

SKILLS

Python; C++, R, HTML,
JS, CSS, Emacs Org
(Lisp), SQL, Matlab.

Linux (Lubuntu+mint);
Nextcloud servers;
Adobe Premiere;
Davinci Resolve

Portuguese; C1 English;
Advanced Spanish;
Intermediate French.

HONORS

- Featured on Portimão Journal for **maximum grades** (20/20) in all courses in high school.
- 5 year degree for music formal education (theory+guitar+chorus)

PROJECTS (independent research + extracurricular)

- **Digital brain** (Coding projects available at github.com/giljorge0):

Built a local-first neuro-symbolic memory system integrating symbolic knowledge graphs, vector retrieval, and temporal reasoning for LLM-assisted querying. Optimized a NetworkX/SQLite architecture to scale beyond 50k nodes using NumPy batched operations and deployed a FastAPI + Ollama backend with LangGraph orchestration. Achieved a 14% MRR improvement over baseline semantic RAG and developed continual-learning mechanisms using Louvain clustering and PageRank-based knowledge-gap detection.

- **Emergent AI-AI communication and geometry of meaning research programme:**

Wrote a theoretical proposal on deriving a universal communication protocol from multilingual embedding spaces, then designed 24 experiments to test pieces of it empirically. Built a zero-knowledge metric for verifying genuine agent communication without decoding message content; mapped ~5,000 concepts across 12 languages using LaBSE to derive speaker-weighted interlingua centroids, translatability tiers and frame-role geometry; quantified a fundamental utility–privacy trade-off in dense AI-to-AI channels that noise and adversarial training cannot resolve. The pipeline spans vocabulary construction, compositional projection, and optimal channel design.

- **Writing** (giljorge0.github.io):

300 thousand written words, on [personal website](https://giljorge0.github.io) preparing selected pieces for publication on public platforms. Mostly philosophy.

- **Multisport athlete** (<https://www.youtube.com/@giljorge7479>):

Compete in university's national track and field championships, represented Sporting Clube de Portugal for 2 years. Also competing in Greco-Roman/freestyle wrestling, recently obtaining 4th place at national championships, and now train in Portugal's high performance training center with the national team.